

Database Tables — Detailed Guide

Updated: 2026-04-17 | Based on actual production DB (5 patients processed)
Total: 10 tables | Branch: `feat/save-intermediate-results`

1. transcript_analysis_log (5 rows)

Pipeline execution record — stores the fact that "this patient file was processed" along with all metadata about the processing.

When the pipeline processes `Input_Keystrokes_REC001 (SID 14).xlsx`, one row is created in this table. It records when processing started (`pipeline_started_at`), how many sentences were analyzed (`total_sentences=424`), what settings were used (`top_n=10`), a backup of the result xlsx (`xlsx_data`), whether the AI pipeline completed (`processed=True`), and the overall consultation quality score (`ai_overall_score=3.67`). Next time the same file is encountered, this table is checked and already-processed files are skipped.

Column	Type	Sample (SID_10)	Description
id	SERIAL PK	1	Unique analysis run ID
patient_id	VARCHAR	SID_10	Patient identifier (extracted from filename)
total_sentences	INT	428	Number of sentences after R stringi segmentation
top_n	INT	10	Top-K sentences selected per domain
context_window	INT	3	±3 surrounding sentences for context
model_results	JSONB	None	(Deprecated — kept for backward compatibility)
xlsx_data	BYTEA	(18,300 bytes)	Result xlsx backup. Fallback when disk file deleted
source_filename	VARCHAR	Input_Keystrokes_REC001 (SID 10).xlsx	Original input filename
pipeline_started_at	TIMESTAMP	05:26:14	When pipeline_runner began processing
analyzed_at	TIMESTAMP	05:26:30	When NLP results saved to DB (Step 8)
ai_overall_score	FLOAT	3.4	Average of all domain ai_scores (0-5). Displayed on DOCTOR page

processed	BOOLEAN	True	True when full pipeline (NLP + AI) completed
processed_at	TIMESTAMP	05:29:28	When AI pipeline (Step 9) completed

Used by:

- `persistence.file_already_processed()` — skip already-processed files
- `GET /api/transcript/download/{patient_id}` — `xlsx` download fallback from `xlsx_data`
- `GET /api/transcript/history/{patient_id}` — analysis run history
- `llm_domain_scoring_and_summary.analysis_id` FK target

2. sentence_prediction (250 rows = 5 patients × 50 sentences)

NLP model predictions — stores the Top-10 sentences per domain with their probability scores, as determined by the 5 NLP classification models.

For example, out of SID 10's 428 sentences, the cp (cancer prognosis) model assigned 95.1% probability to "so i'm going to take that 12 percent and cut it in half...", making it one of the Top-10 for that domain. 5 domains × 10 sentences = 50 rows per patient.

Where this appears in the UI: In `PatientInitialVisitReportV35.tsx`, each `TopicCard` has a collapsible "Evidence" section. When the patient clicks "Show Evidence", it displays a list of sentences ranked by `pred_score`. These sentences come from this table via `GET /api/patient/sentences/{file}` (Top-7 per domain).

Column	Type	Sample (SID_10, cp)	Description
id	SERIAL PK	1	Unique prediction ID
analysis_id	INT FK	1	→ <code>transcript_analysis_log.id</code>
patient_id	VARCHAR	SID_10	Patient identifier
model	VARCHAR	cp	Domain (cp/le/ed/inc/ius)
sentence_index	INT	166	Global sentence sequence number
utterance_index	INT	67	Original utterance number (i)
sentence_in_utterance	INT	3	Sentence number within utterance (i2)
speaker	VARCHAR	Interviewer:	Speaker
sentence_text	TEXT	"so i'm going to take that 12 percent and cut it in half..."	Original sentence
pred_score	FLOAT	0.951	NLP probability (95.1% related to cp)

context	TEXT	"it removes the testosterone...<main>so i'm going to...</main>..."	±3 surrounding sentences
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Used by:

- GET /api/patient/sentences/{file} — evidence sentences on PATIENT page
- Analysis result reconstruction from DB

3. doctor_sentence_view (221 rows)

Doctor dashboard sentence list — stores deduplicated sentences for the doctor to review and optionally rewrite. Unlike sentence_prediction where the same sentence can appear in multiple domains, here each sentence appears only once with one representative domain.

The reason this table has 221 rows (less than sentence_prediction's 250) is deduplication: if the same sentence ranked in Top-10 for both cp and le domains, it appears in sentence_prediction twice but in doctor_sentence_view only once. The doctor sees each sentence exactly once. When the doctor rewrites a sentence, (file, i, i2) from this table identifies the target.

Where this appears in the UI: In PhysicianReportsModifiedV41Timothy.tsx, the **GridView** displays a table with columns: "Topic", "Your Score", "Representative Sentence", "Suggestions for Improvement", and "Suggested Rephrasing". The "Your Score" comes from llm_domain_scoring_and_summary.ai_score (via scores/summary API). The "Representative Sentence" shows the full context (±3 sentences) from sentence_prediction.context (via sentences API), with the key sentence highlighted in bold+underline. Clicking a topic opens the **DetailView** with Consultation Scoring bubble and Re-write Practice, which also display context from the same sentences API.

Column	Type	Sample	Description
file	VARCHAR PK	Input_Keystrokes REC 001 (SID 10).xlsx	Patient file
i	INT PK	67	Utterance number
i2	INT PK	3	Sentence within utterance
speaker	VARCHAR	Interviewer:	Speaker
sentence	TEXT	"so i'm going to take that 12 percent..."	Sentence text
score	FLOAT	None	Not used directly. AI score is read from llm_domain_scoring_and_summary.ai_score via JOIN in the sentences API. This column remains NULL.
class	VARCHAR	cancer_prognosis	Representative domain
time	TIMESTAMP	05:26:30	Creation time

Used by:

- GET /api/doctor/sentences/{file}/{speaker} — DOCTOR page sentence list
- GET /api/doctor/files — processed patient file list (DISTINCT file)
- PUT /api/doctor/rewrites — rewrite target identification via (file, i, i2)
- GET /api/patient/sentences/{file} — evidence sentences on PATIENT page

4. doctor_rewrite_log (0 rows)

Doctor rewrite history — records every time a doctor modifies a sentence on the dashboard, storing both the original and revised versions with timestamps.

Currently 0 rows because no doctor has rewritten any sentences yet. When a doctor changes "so i'm going to take that 12 percent..." to "Your cancer risk is about 6% with treatment", both versions are INSERT'd here. Multiple rewrites of the same sentence create multiple rows, preserving the full edit history.

Where this appears in the UI: In `PhysicianReportsModifiedV41Timothy.tsx`, the **GridView** has a "Suggested Rephrasing" column. When the doctor clicks "AI Rewrite", GPT-4o generates an improved version. If the doctor accepts and saves, the original and revised sentences are stored here. The **HistoryModal** component shows the full rewrite history for a selected sentence.

Column	Type	Example	Description
file	VARCHAR PK,FK	Input_Keystrokes REC 001 (SID 10).xlsx	Target file
i	INT PK,FK	67	Target utterance number
i2	INT PK,FK	3	Target sentence number
time	TIMESTAMP PK	2026-04-17T14:30:00Z	Rewrite time (multiple rewrites = multiple rows)
speaker	VARCHAR	Interviewer:	Speaker
original_sentence	TEXT	"so i'm going to take..."	Original sentence
revised_sentence	TEXT	"Your cancer risk is about 6%..."	Doctor's rewritten version
score	FLOAT	4.0	Quality score after rewrite
class	VARCHAR	cancer_prognosis	Domain

Used by:

- GET /api/doctor/rewrites — rewrite history on DOCTOR page
- GET /api/doctor/rewrites/{file}/{i}/{i2}/history — per-sentence rewrite history

5. patient_summary (5 rows)

Patient summary parent record — represents "a summary exists for this patient". Acts as the FK parent for `patient_summary_domain` (5 domain rows per patient).

One row per patient. The actual domain-level data (`summary_text`, `patient_scoring`) lives in the child table `patient_summary_domain`. This table exists because DB design requires a "1" side table in a 1:N

relationship for FK integrity. Deleting a patient here CASCADE-deletes all 5 domain rows.

Column	Type	Sample	Description
file	VARCHAR PK	Input_Keystrokes REC 001 (SID 10).xlsx	Patient file
speaker	VARCHAR PK	Patient_Input_Keystrokes REC 001 (SID 10)	Patient identifier
entire_summary	TEXT	None	Full visit summary (currently unused)

Used by:

- GET /api/patient/summaries/{file}/{speaker} — JOINS with patient_summary_domain
- FK parent for patient_summary_domain CASCADE delete

6. patient_summary_domain (25 rows = 5 patients × 5 domains)

Patient per-domain feedback — stores the star rating and text response that the patient enters on the dashboard for each domain. The pipeline creates empty rows (NULL values), and the patient fills them in later during a follow-up visit.

Like a patient survey form: the pipeline creates 5 blank domain cards (cp, le, ed, inc, ius), and when the patient clicks a rating for "cancer prognosis", `patient_scoring` is UPDATE'd from NULL to the rating value. Note: `patient_response` (free text) has an API but no UI implementation yet.

Where this appears in the UI: In `PatientInitialVisitReportV35.tsx`, each **TopicCard** has a NIH PROMIS unipolar helpfulness scale (5 levels: "Not at all helpful" to "Extremely helpful") at the bottom. When the patient selects a level, `updateSingleClassScore()` is called → `PUT /api/patient/scoring` → this table's `patient_scoring` column is UPDATE'd. The `display_order` determines the vertical order of the 5 topic cards on the page (1=Cancer Prognosis at top, 5=Life Expectancy at bottom).

Column	Type	Sample (SID_10, cp)	Description
file	VARCHAR PK,FK	Input_Keystrokes REC 001 (SID 10).xlsx	Patient file
speaker	VARCHAR PK,FK	Patient_...SID 10	Patient identifier
domain	VARCHAR PK	cancer_prognosis	Domain
display_order	INT	1	UI display order (1=cp, 2=inc, 3=ed, 4=ius, 5=le)
summary_text	TEXT	""	Domain summary text (currently empty)
patient_scoring	INT	None	Patient star rating 0-10 (NULL until patient rates)

patient_response	TEXT	None	Free text feedback (API exists but UI NOT implemented)
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Used by:

- GET /api/patient/summaries/{file}/{speaker} — domain summary cards on PATIENT page
- PUT /api/patient/scoring — patient star rating (PatientInitialVisitReportV35.tsx)
- PUT /api/patient/responses — patient text response (API only, UI not implemented)

7. survey_submission_log (0 rows)

Patient survey responses — stores the full JSON answers when a patient submits SDM, DCS, Risk Perception, or Satisfaction surveys on the follow-up page, and tracks synchronization status with the REDCap research database.

Currently 0 rows because no patient has submitted a survey yet. When a patient completes the DCS (16 questions), the entire response is stored as JSON in `answers`. Immediately after, a background task calls the REDCap API to sync. Same patient submitting the same survey again creates a NEW row (INSERT, not UPDATE) to preserve responses at different timepoints.

Where this appears in the UI: In `PatientFollowUpReportV31Re.tsx`, the follow-up visit has a multi-step survey flow: Welcome → SDM (9 questions) → DCS (16 questions) → Risk Perception (with collapsible AI summaries) → Patient Satisfaction → Completion. Each survey shows one question at a time. When the patient clicks "Submit", `submitSurvey()` from `surveyApi.tsx` calls `POST /api/surveys/submit`, which INSERTs a row here. On page refresh, `fetchSurveySubmissions()` restores previous answers so the patient can continue where they left off.

Column	Type	Example	Description
id	SERIAL PK	1	Submission ID
file	VARCHAR FK	Input_Keystrokes REC001 (SID 14).xlsx	Patient file
speaker	VARCHAR FK	Patient_...SID 14	Patient identifier
survey_type	VARCHAR	dcs	Survey type (sdm/dcs/risk_perception/satisfaction)
answers	JSONB	{q1: 2, q2: 3, ..., q16: 4}	Full survey response
extra_data	JSONB	{browser: Chrome, session: abc}	Metadata
submitted_at	TIMESTAMP	2026-04-17T16:05:30Z	Submission time
redcap_synced	BOOLEAN	true	REDCap sync success
redcap_record_id	VARCHAR	REC-2026-0014	REDCap record ID (on success)

redcap_error	TEXT	None	Error message (on failure)
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Used by:

- `POST /api/surveys/submit` — survey submission from PATIENT follow-up page
- `GET /api/surveys/by-speaker/{speaker}` — restore previous responses on page refresh
- Background REDCap sync worker

8. llm_domain_scoring_and_summary (33 rows)

GPT-4o AI pipeline results — stores the per-domain evaluation of "how specifically did the doctor communicate risk to the patient", including a 0-5 score, extracted numerical estimates, and a patient-friendly reformatted summary.

While NLP (sentence_prediction) determines "which sentences are related to this domain", this table stores "how specifically did the doctor communicate risk in those sentences". Example: "There's some cancer risk" = ai_score 1 (vague), "24% risk without treatment, decreases to 6% with surgery" = ai_score 4 (very specific). The 33 rows (more than 5×5=25) occur because some domains have multiple treatment comparisons stored as separate rows.

Where this appears in the UI (two different pages, two different columns):

1. **Doctor page** (PhysicianReportsModifiedV41Timothy.tsx):

- **DashboardView:** The patient list shows each patient's overallScore (from ai_overall_score in transcript_analysis_log , which is the average of this table's ai_score). Patients are color-coded: green (score ≥ 4), yellow (2-3), red (< 2).
- **GridView:** The "Your Score" column shows per-domain ai_score (0-5) for the selected patient. The **DetailView** shows a Recharts LineChart trajectory of ai_score over time via GET /api/doctor/scores/trajectory .

2. **Patient page** (PatientInitialVisitReportV35.tsx and PatientFollowUpReportV31Re.tsx):

- Each **TopicCard** displays the reformat_sentence as the main AI summary text. For example, the Cancer Prognosis card shows: "Your doctor noted that, without treatment, your risk of dying of cancer is 24–25%. With treatment, this risk decreases to 6%." This text comes from GET /api/patient/ai-summary/{file} → this table's reformat_sentence column.

Column	Type	Sample (SID_10, cp)	Description
id	SERIAL PK	1	Result ID
analysis_id	INT FK	1	→ transcript_analysis_log.id
patient_id	VARCHAR	SID_10	Patient identifier
domain	VARCHAR	cp	Domain
ai_score	INT	4	GPT-4o score 0-5 → DOCTOR page

score_explanation	TEXT	"Does the text mention cancer mortality?... score is 4."	GPT-4o chain-of-thought reasoning (from ai_pipeline/scoring.py)
extracted_estimate	TEXT	"24, 25 percent—down to six percent"	Extracted risk numbers
treatment	VARCHAR	None	Related treatment (ed/inc/ius only)
source_sentence	TEXT	"so i'm going to take that 12 percent..."	AI-selected original sentence
source_context	TEXT	"it removes the testosterone...<main>...</main>..."	Surrounding context
reformat_sentence	TEXT	"Your doctor noted that your risk of dying of cancer is 24–25%..."	Patient-friendly text → PATIENT page
source_filename	VARCHAR	Input_Keystrokes REC 001 (SID 10).xlsx	Source file
created_at	TIMESTAMP	05:29:28	Creation time

Used by:

- GET /api/doctor/scores/average — consultation quality score on DOCTOR page
- GET /api/doctor/scores/trajectory — score trend over time on DOCTOR page
- GET /api/patient/ai-summary/{file} — AI summary card on PATIENT page
- POST /api/doctor/ai-rewrite — AI rewrite suggestions using source_sentence + source_context

9. user_interaction_log (108 rows)

User behavior tracking — records every user action on the dashboard (clicks, scrolls, mouse movements, dwell time) in real-time for research analysis of how patients and doctors interact with the consultation information.

For example, when a patient hovers near the "cancer prognosis" card for 38 seconds, an event is recorded with event_type=cursor_proximity_leave and hoverDuration: 38007071ms . This data enables research questions like "Which domain information do patients spend the most time reading?" or "Which sentences do doctors click on most?"

Where this appears in the UI: This table is NOT visible to patients or doctors. It is consumed by the **AdminTrackingDashboard** (AdminTrackingDashboard.tsx), which shows event counts, session timelines, per-patient behavior heatmaps, and event type distributions. Accessible only to administrators.

Column	Type	Sample	Description
id	SERIAL PK	1	Event ID

session_id	VARCHAR	session_1776435900666_syzu2gq6lrk	Browser session ID
role	VARCHAR	patient	User role (patient/physician)
visit_type	VARCHAR	first	Visit type (first/followup)
file	VARCHAR	Input_TurboScribe SID 33.csv	Patient being viewed
speaker	VARCHAR	Patient_Input_TurboScribe SID 33	Patient identifier
event_type	VARCHAR	cursor_proximity_leave	Event type
element_id	VARCHAR	/	Target UI element
event_data	JSONB	{cursorX: 0, hoverDuration: 38007071, ...}	Event details
device_type	VARCHAR	desktop	Device type
client_timestamp	TIMESTAMP	14:25:00	Client-side time
created_at	TIMESTAMP	14:25:00	Server save time

Used by:

- `POST /api/tracking/events` — event batch submission from frontend
- `GET /api/tracking/stats` — admin tracking dashboard
- `GET /api/tracking/analytics` — behavior analysis

10. session_recording (2 rows)

Session replay data — stores rrweb-recorded user sessions as binary chunks, enabling video-like playback of how users navigated the dashboard. PHI (Protected Health Information) is masked so patient names and sentence text appear as "" in recordings.*

While `user_interaction_log` records individual events, this table stores the raw data needed to replay the entire session visually. Chunks are sent every 30 seconds or every 500 events.

Where this appears in the UI: In the **AdminTrackingDashboard** (`AdminTrackingDashboard.tsx`), the "Recordings" tab lists all recorded sessions. Clicking a session opens the **rrweb-player** component, which plays back the session as a video showing mouse movements, clicks, scrolls, and page content (with PHI masked as ""). Not visible to patients or doctors.

Column	Type	Sample	Description
id	SERIAL PK	1	Recording ID
session_id	VARCHAR	rec_1776397890521_hp3r5x	Recording session ID
chunk_index	INT	0	Chunk number (split every 30s or 500 events)

file	VARCHAR	Input_TurboScribe SID 33.csv	Patient being viewed
visit_type	VARCHAR	first	Visit type
recording_data	BYTEA	(87 bytes)	rrweb event data (PHI masked)
event_count	INT	2	Events in this chunk
created_at	TIMESTAMP	14:25:30	Server save time

Used by:

- `POST /api/tracking/recordings` — chunk upload from frontend (sessionRecorder.ts)
- `GET /api/tracking/recordings` — recording list on admin dashboard
- `GET /api/tracking/recordings/{sessionId}` — session replay playback